Danger Is a Counterfactual: Toward a Falsifiable Continuous Test Oracle for Autonomous Driving Systems

Borhaneddine Hamadou
Universite Paris-Saclay, CEA-List
Palaiseau, France
borhaneddine.hamadou@cea.fr

Abstract

The verdict that decides a simulated autonomous-driving test is a single bit: collision or no collision. The bit calls a four-centimetre miss at highway speed a pass, and a change that makes crashes more violent — but no more frequent — never changes it, at any testing volume. Continuous danger scores exist; we are aware of none validated on the run it judges, because a near-miss ends without injury, leaving no outcome to check a score against. We propose to measure rather than score: the harm of a test run is the fraction of its plausible neighbouring worlds — slower reactions, weaker brakes, jittered traffic, re-simulated with the system reacting freshly in each — that end in a crash, weighted by severity. Ground truth becomes re-simulation, and a danger estimate can finally be *wrong*. The measure sees what the verdict cannot: one run in our openpilot suite is classified as a pass by every telemetry-based baseline, yet 70 of its 100 disturbed replays end in a 72 km/h head-on crash — a third of the entire suite's harm, hidden in a single pass. A cheap estimate distilled from the reference campaign, computed from the recorded run alone, surfaces 61% of total suite harm under a budget permitting inspection of 15 of the 150 runs (best telemetry heuristic: 35%), matches three-replay measurement in ranking accuracy at a quarter of its cost ($r_s = 0.75$), and reports calibrated uncertainty: 94% of its nominal 90% intervals contain the truth.

CCS Concepts

• **Software and its engineering** → **Software testing and debugging**; • **Computer systems organization** → *Embedded and cyber-physical systems.*

Keywords

test oracle, autonomous driving, scenario-based testing, counterfactual analysis, simulation

1 Problem and Motivation

Autonomous driving systems cannot be validated by road testing alone [9], so they are validated by scenario-based testing: concrete traffic situations are executed in a simulator and judged by a test oracle. A decade of research has optimised nearly every stage of that pipeline — generation, selection, prioritisation — while the oracle that decides what each execution *means* often uses criteria such as a collision/no-collision requirement [2], a time-to-collision threshold [6], or a safety-envelope check [16]. One run, one bit.

The bit discards the property a safety engineer actually cares about. Two executions of an overtaking scenario — one passes with five metres to spare at low speed, the other misses a truck by four centimetres at 100 km/h — get the same verdict: pass. And the loss compounds: a planner change after which the vehicle crashes exactly as often as before, but harder, never changes the distribution of the bit, so no volume of testing can detect it. In our pilot experiments, statistical tests on the binary verdict detected planted defects of exactly this kind with power 0.03–0.09 — chance level — at every budget tried.

2 Why the Obvious Fix Fails

The remedy seems obvious: report a continuous score — time-to-collision (TTC), clearance, or any of dozens of criticality metrics [8, 18]. Two obstacles, one empirical and one methodological, explain why testing still runs on one bit.

First, **proximity is not danger**: the gap characterises the world that occurred, not how the system would have behaved in nearby ones. In our campaign (§5), a run that TTC ranks among the most critical in the suite (0.45 s, an oncoming vehicle) never crashes in 100 disturbed re-simulations — the ego was at walking pace, in control; a run that TTC rates five times safer (2.3 s) rear-ends a stopped vehicle at 40 km/h in one disturbed replay in ten. TTC orders the pair backwards: it cannot tell control from luck.

Second, **the judged run offers nothing to validate a score against**. Suppose an oracle assigns the four-centimetre miss a danger of 0.7. Is that correct? The run ended without injury, so the observed outcome contains nothing to check the number against; asking experts only tests whether the score imitates intuition about the very telemetry it was computed from. A verdict no observation can contradict is a scoring convention, not a measurement. Model-based collision probabilities are catalogued in [18]; for a run that ends without a crash, their outputs, too, cannot be checked against that run alone.

3 Measure Danger by Replaying It

Why do we all agree the four-centimetre miss was dangerous, when nothing bad happened? Because it *almost* happened: a slightly slower reaction or a wetter road ends it in a violent crash. Danger is a property of the neighbourhood of worlds around a run — and scenario-based testing lives in the one place where those worlds are computable. So we measure them:

$$H(e) \;=\; \frac{1}{N}\sum_{i=1}^{N} \iota(s_i), \tag{1}$$

where e is the run under judgment — one executed scenario, recorded as a telemetry trace; each of the N replays re-executes e under one plausible disturbance — reaction onset delayed by up to ±0.25 s, braking at 80–105% strength, small random jitter on the other party's motion, drawn from a declared kernel κ — with the system under test reacting freshly inside every replay; s_i is the impact speed

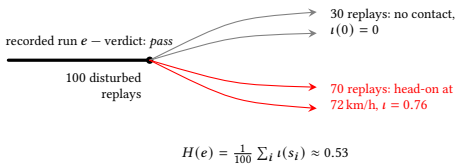


Figure 1: The measure on a real suite run: an oncoming car drifts into the ego lane and openpilot squeezes past with 1.4 m to spare. The verdict says pass; the neighbourhood says 70% of plausible worlds end in a 72 km/h head-on. This one run carries a third of the total harm of the 150-run suite.

of replay i (relative speed at contact, 0 if the replay ends without contact); and ι is our severity function: a logistic curve inspired by injury-risk models [11], mapping impact speed to a weight between 0 and 1, with $\iota(0) = 0$. Like κ, its coefficients are declared parameters of the oracle. $H(e)$ is the severity-weighted fraction of the run's plausible neighbourhood that ends in a crash. Counterfactual re-simulation of this kind is established in safety-impact estimation, where recorded crashes and conflicts are replayed to credit a system with the crashes it avoids ("fractional collisions") [1, 15]; we invert the construction and use it as the ground truth of a test oracle.

Figure 1 shows the measure at work on the run whose pass verdict is most misleading. Three properties make H a foundation rather than another score. It is *falsifiable*: "the harm of this run is 0.53" is confirmed or refuted by more replays — the property the scores of §2 lack. It *subsumes the verdict*: shrink the disturbances to zero and make ι a step function, and H is exactly the collision bit. And the disturbance model becomes an explicit, auditable parameter, analogous to the controllability parameter of ISO 26262-3 [7]: harm is reported kernel-indexed.

4 A Single-Run Estimate

Replays cost simulation time. The practical oracle is two-tier: a reference tier measures H by replay where it matters, and a low-cost estimator predicts it from the recorded run alone, requiring no simulation beyond the nominal run that testing already executes. Our estimator rank-fuses three signals, all computed from the single nominal trace: gradient-boosted trees trained on the reference campaign's 15,000 *individual* replay outcomes, each one a (run features, disturbance, crashed?) example, integrated over the kernel at prediction time; forests fit to the scenario-level harm labels on a log scale; and a zero-parameter physics score — ι(closing speed at the most critical moment) $/(1 +$ closest clearance), the definition's own structure in closed form — which flags rare catastrophic patterns too scarce to learn (Figure 1's run is one). The magnitude estimate carries a split-conformal interval [12].

5 Validation and Results

Because ground truth is computed rather than elicited, validation needs no human judgment: predictions are compared against re-simulated truth with standard instruments — interval coverage for calibration [5], rank correlation for ordering [17]. Subject: openpilot [3], an open-source advanced driver-assistance system, in MetaDrive [13]; 150 concrete scenarios from 5 NHTSA-style pre-crash templates [14], each replayed M=100 times under the declared kernel (15,000 reference simulations). Splits are at the scenario level, never the trace level.

Table 1: Share of the suite's total harm in the 15 runs (10%) each ordering ranks first, and rank correlation r_s with re-simulated true harm over all 150 runs. Model rows are out-of-fold — every run scored by models that never saw it (10× five-fold by scenario, mean ± s.d.); replay row: 400 subsample draws; binary-verdict ties: broken at random (harm share), mid-ranked (r_s).

Ordering of runs	Sim. cost	Harm surfaced	r_s
Binary collision verdict	1×	26.2% ± 3.6	0.64
Minimum time-to-collision	1×	34.5%	0.40
Predicted harm *(this work)*	1×	**61.5% ± 2.2**	**0.75 ± 0.01**
Measured harm, 3 replays	4×	76.5% ± 5.7	0.75 ± 0.04
True harm (M=100 replays)	101×	83.3%	1

It finds the danger the bit ignores. Suppose a tester can hand-inspect only 15 of the 150 runs and picks them by each score. Table 1: the 15 that predicted harm ranks first contain 61.5% of the suite's total harm, at single-run cost — 1.8× the best telemetry heuristic. Ordering is stronger still: the estimate ranks runs as well as three-replay measurement ($r_s = 0.75$ for both), making it the natural screen for deciding where replays are worth spending.

It orders exactly the cases the bit cannot. Among held-out non-collision runs — all judged identically by the verdict — predicted harm ranks re-simulated truth at Spearman $r_s = 0.60 \pm 0.02$ (0.75 ± 0.01 over all runs). The verdict returns identical judgments for 22% of held-out pairs whose true harms differ by more than 10×; a TTC threshold, 24%; the estimate, none.

The uncertainty is calibrated. Every prediction is issued with an interval constructed — by conformal calibration on held-out runs — to contain the re-simulated truth 90% of the time. On test runs it does: the truth landed inside 94% of the intervals, so the stated confidence is kept, erring slightly safe. The reference labels are themselves precise: their Monte-Carlo error is 7× below the model's error.

And it detects what the bit cannot. In the pilot study's planted severity regressions (crash rate fixed, crashes more violent), binary-verdict power stayed at 0.03–0.09 while the harm measure detected every defect with power 0.97–1.00 at eighty scenarios.

6 Conclusion and Future Work

What blocked replacing the collision bit was never a shortage of scores: proximity characterises the observed world, not its fragile neighbours, and a run that ends without injury offers no observation to validate a score against. Counterfactual harm dissolves both obstacles and subsumes the verdict. This paper contributes the falsifiable definition with the kernel a declared parameter; a validation methodology reusable wherever verdicts are graded and single outcomes censored; and evidence on a real driver-assistance system that a distilled single-run estimate surfaces most of the danger the verdict ignores. The measure is a one-off investment: the reference campaign cost roughly 800 GPU-hours; once distilled, scoring a new run takes milliseconds, and replays are reserved for confirming what the estimate escalates. What a test oracle for a stochastic safety-critical system should report is not whether the observed world went wrong, but how much of the adjacent worlds do.

Future work. A seed-paired power study of planted defects is in execution: can the estimate detect the regressions the bit cannot, and at what budget? Replication on Autoware in CARLA [4, 10] follows.

References

[1] Jonas Bärgman, Christian-Nils Boda, and Marco Dozza. 2017. Counterfactual simulations applied to SHRP2 crashes: The effect of driver behavior models on safety benefit estimations of intelligent safety systems. *Accident Analysis & Prevention* 102 (2017), 165–180.

[2] Raja Ben Abdessalem, Annibale Panichella, Shiva Nejati, Lionel C. Briand, and Thomas Stifter. 2018. Testing Autonomous Cars for Feature Interaction Failures using Many-Objective Search. In *2018 33rd IEEE/ACM International Conference on Automated Software Engineering (ASE)*. 143–154. doi:10.1145/3238147.3238192

[3] comma.ai. 2026. openpilot: An open source driver assistance system. https://github.com/commaai/openpilot, commit `492ed731` (master, 2026-05-11).

[4] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. 2017. CARLA: An Open Urban Driving Simulator. In *Proceedings of the 1st Annual Conference on Robot Learning (Proceedings of Machine Learning Research, Vol. 78)*, Sergey Levine, Vincent Vanhoucke, and Ken Goldberg (Eds.). PMLR, 1–16. https://proceedings.mlr.press/v78/dosovitskiy17a.html

[5] Tilmann Gneiting and Adrian E Raftery. 2007. Strictly proper scoring rules, prediction, and estimation. *Journal of the American statistical Association* 102, 477 (2007), 359–378.

[6] John C. Hayward. 1972. Near-miss determination through use of a scale of danger. *Highway Research Record* 384 (1972), 24–34.

[7] International Organization for Standardization. 2018. ISO 26262-3:2018: Road vehicles — Functional safety — Part 3: Concept phase.

[8] Gunel Jahangirova, Andrea Stocco, and Paolo Tonella. 2021. Quality metrics and oracles for autonomous vehicles testing. In *2021 14th IEEE conference on software testing, verification and validation (ICST)*. IEEE, 194–204.

[9] Nidhi Kalra and Susan M Paddock. 2016. Driving to safety: How many miles of driving would it take to demonstrate autonomous vehicle reliability? *Transportation research part A: policy and practice* 94 (2016), 182–193.

[10] Shinpei Kato, Shota Tokunaga, Yuya Maruyama, Seiya Maeda, Manato Hirabayashi, Yuki Kitsukawa, Abraham Monrroy, Tomohito Ando, Yusuke Fujii, and Takuya Azumi. 2018. Autoware on board: Enabling autonomous vehicles with embedded systems. In *2018 ACM/IEEE 9th International Conference on Cyber-Physical Systems (ICCPS)*. IEEE, 287–296.

[11] Kristofer D Kusano and Hampton C Gabler. 2012. Safety benefits of forward collision warning, brake assist, and autonomous braking systems in rear-end collisions. *IEEE Transactions on Intelligent Transportation Systems* 13, 4 (2012), 1546–1555.

[12] Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. 2018. Distribution-free predictive inference for regression. *J. Amer. Statist. Assoc.* 113, 523 (2018), 1094–1111.

[13] Quanyi Li, Zhenghao Peng, Lan Feng, Qihang Zhang, Zhenghai Xue, and Bolei Zhou. 2023. MetaDrive: Composing Diverse Driving Scenarios for Generalizable Reinforcement Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 3 (2023), 3461–3475. doi:10.1109/TPAMI.2022.3190471

[14] Wassim G. Najm, John D. Smith, and Mikio Yanagisawa. 2007. *Pre-crash scenario typology for crash avoidance research.* Technical Report DOT HS 810 767. National Highway Traffic Safety Administration.

[15] Sreeja Roy-Singh, Sarvesh Kolekar, Daniel P Bonny, and Kyle Foss. 2025. Fractional Collisions: A Framework for Risk Estimation of Counterfactual Conflicts using Autonomous Driving Behavior Simulations. *arXiv preprint arXiv:2506.07540* (2025).

[16] Shai Shalev-Shwartz, Shaked Shammah, and Amnon Shashua. 2017. On a formal model of safe and scalable self-driving cars. *arXiv preprint arXiv:1708.06374* (2017).

[17] Charles Spearman. 1904. The proof and measurement of association between two things. *The American Journal of Psychology* 15, 1 (1904), 72–101.

[18] Lukas Westhofen, Christian Neurohr, Tjark Koopmann, Martin Butz, Barbara Schütt, Fabian Utesch, Birte Neurohr, Christian Gutenkunst, and Eckard Böde. 2023. Criticality Metrics for Automated Driving: A Review and Suitability Analysis of the State of the Art. *Archives of Computational Methods in Engineering* 30, 1 (Jan. 2023), 1–35. doi:10.1007/s11831-022-09788-7